Study on the Positive Correlation Between Fish Culture Duration and the Quantity of Edible Parts

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Date: 2026

Abstract

Objective: This study aims to investigate the positive correlation between fish culture duration and the quantity of edible parts, providing scientific evidence for determining optimal harvest times in aquaculture.

Methods: Based on a systematic review of recent research on Nile tilapia (Oreochromis niloticus), rainbow trout (Oncorhynchus mykiss), Chinese softshelled turtle (Pelodiscus sinensis), and Chinese mitten crab (Eriocheir sinensis), this paper analyzes the dynamic changes in fillet yield, meat quality, protein content, and fatty acid composition across different culture durations.

Results: The findings reveal that: (1) Fillet weight and carcass weight in Nile tilapia show strong positive correlations with culture duration (genetic correlations >0.98 between body weight and fillet weight) ; (2) Protein content and EPA+DHA levels in Chinese softshelled turtle increase progressively with age, reaching optimal levels at 6 years ; (3) The relationship between culture duration and edible parts follows a "diminishing returns" pattern, where growth rate declines after sexual maturity while nutritional quality may continue to improve; (4) Excessive culture duration (beyond 60 days in overwintering crabs) leads to decreased edible yield and nutritional quality .

Conclusion: A positive correlation exists between fish culture duration and edible parts quantity, but this relationship is nonlinear and speciesspecific. The optimal harvest time should balance yield quantity, nutritional quality, and economic efficiency.

Keywords: fish culture duration; edible parts; fillet yield; nutritional quality; aquaculture optimization

1. Introduction

Aquaculture has become the fastestgrowing food production sector globally, supplying over 50% of fish for human consumption. Among the critical decisions facing aquaculturists is determining the optimal harvest time for farmed fish.

Traditional practices often rely on the assumption that "longer culture duration produces better fish," yet this belief lacks systematic scientific validation.

The edible parts of fish—primarily muscle tissue, but also including other consumable organs—represent the economic core of aquaculture production. Understanding how these edible portions change quantitatively and qualitatively throughout the culture cycle is essential for maximizing both yield and profitability.

This study addresses three fundamental questions: (1) Does prolonged culture duration consistently increase the quantity of edible parts? (2) What physiological mechanisms govern the relationship between age and edible tissue accumulation? (3) How can farmers determine the optimal harvest time that balances yield quantity, nutritional quality, and production costs?

By synthesizing recent research across multiple aquaculture species, this paper aims to establish a scientific framework for understanding the correlation between culture duration and edible parts quantity, ultimately contributing to more efficient and sustainable aquaculture practices.

2. Materials and Methods

2.1 Literature Search Strategy

A comprehensive literature search was conducted using academic databases including Web of Science, Google Scholar, and China National Knowledge Infrastructure (CNKI). Search terms included combinations of: "fish culture duration," "culture time," "age," "fillet yield," "edible parts," "meat yield," "nutritional quality," and specific species names. Priority was given to peerreviewed studies published within the last 15 years that provided quantitative data on the relationship between culture duration and edible tissue parameters.

2.2 Data Extraction and Analysis

Key data extracted from selected studies included:

Species and culture conditions

Culture duration or age ranges examined

Body weight and fillet weight measurements

Proximate composition (protein, lipid, moisture, ash)

Fatty acid profiles, particularly EPA and DHA content

Amino acid composition

Yield indices (fillet yield, dressout percentage)

2.3 Statistical Considerations

Where available, correlation coefficients, heritability estimates, and significance levels were recorded. For studies presenting longitudinal data, trends across time points were analyzed to identify patterns of increase, plateau, or decline in edible parts quantity and quality.

3. Results

3.1 Body Weight and Fillet Weight Relationships

Longitudinal genetic analyses of Nile tilapia (Oreochromis niloticus) from 106 to 245 days posthatching revealed strong positive correlations between body weight and both fillet weight and carcass weight . As shown in Figure 1, the genetic correlations exceeded 0.98 when both ages were greater than 125 days, indicating that selection for increased body weight would result in proportional increases in edible fillet yield.

Flavor amino acids peaked at 5 years, suggesting optimal eating quality precedes maximum nutritional density

3.3 Temporal Patterns in Edible Tissue Accumulation

Studies on large rainbow trout (Oncorhynchus mykiss) cultured in Black Sea net cages revealed that growth parameters including weight gain and specific growth rate were highest in groups receiving optimal feeding regimes . The feed conversion ratio

(FCR) ranged from 1.57 to 1.59 in wellfed groups, indicating efficient conversion of feed into edible tissue.

For Chinese mitten crabs (Eriocheir sinensis) subjected to overwintering culture, a more complex pattern emerged .

Notably, while body weight and meat yield declined throughout the 90day period, nutritional quality parameters showed an initial improvement followed by deterioration. Polyunsaturated fatty acid (PUFA) and EPA+DHA contents in hepatopancreas and muscle increased significantly during the first 60 days before declining thereafter .

3.4 SpeciesSpecific Patterns

The relationship between culture duration and edible parts quantity varies considerably among species.

This model illustrates that:

Warmwater fish (tilapia) show sustained increases in edible parts during optimal growing periods

Coldwater fish (trout) exhibit more seasonal patterns with growth slowing during winter

Crustaceans (crabs) may show edible tissue mobilization during reproductive periods, reducing meat yield

Reptiles (turtle) demonstrate continued nutritional improvement even after somatic growth slows

4. Discussion

4.1 Mechanisms Underlying the DurationEdible Parts Relationship

The positive correlation between fish culture duration and edible parts quantity operates through several physiological mechanisms:

Somatic growth and muscle hyperplasia/hypertrophy: In most fish species, muscle tissue continues to accumulate through both an increase in muscle fiber number (hyperplasia) and fiber size (hypertrophy) well beyond sexual maturity . This accounts for the strong correlation between age and fillet weight observed in tilapia.

Nutrient partitioning and storage: As fish age, they develop more efficient mechanisms for nutrient storage. The progressive increase in protein and EPA+DHA content in 6yearold turtles compared to younger individuals reflects enhanced capacity for nutrient assimilation and retention .

Reproductive maturation and tissue mobilization: The decline in edible parts quantity observed in crabs after 60 days of overwintering coincides with gonad development . During reproductive maturation, nutrients are mobilized from somatic

tissues (muscle, hepatopancreas) to support gamete production, temporarily reducing edible yield.

4.2 The "Diminishing Returns" Pattern

The reviewed studies consistently reveal a "diminishing returns" pattern in the relationship between culture duration and edible parts accumulation. This pattern has three phases:

Phase 1 Accelerating growth (juvenile stage): During early development, fish exhibit high growth rates and efficient feed conversion. Edible parts accumulate rapidly, and the marginal gain per unit time is high.

Phase 2 Linear growth (preadult stage): Growth rates stabilize, and edible parts accumulate at a relatively constant rate. This phase often represents the most economically efficient production period.

Phase 3 Decelerating growth (adult/senescent stage): Growth rates decline as fish approach their asymptotic size. Feed conversion efficiency decreases, and the cost of producing additional edible tissue increases. In some species, nutritional quality may continue improving even as growth slows , but in others, tissue mobilization for reproduction may actually reduce edible yield .

4.3 Implications for Aquaculture Management

The findings of this study have several practical implications for aquaculture operations:

Optimal harvest timing: For most species, the optimal harvest time occurs at the transition between Phases 2 and 3—when somatic growth begins to slow but before significant reproductive tissue development or nutritional decline occurs. For Nile tilapia, this occurs around 200245 days posthatching ; for overwintering crabs, within 60 days of holding .

Market segmentation opportunities: The discovery that 5yearold turtles have optimal flavor while 6yearolds have optimal nutrition suggests opportunities for market segmentation based on consumer preferences. Similarly, the improved fatty acid profiles during early overwintering could support premium pricing for crabs harvested at specific times.

Speciesspecific strategies: The variability in durationedible parts relationships across species underscores the need for speciesspecific management protocols. Warmwater fish may benefit from extended culture periods in tropical regions, while temperate species require careful timing to avoid winter weight loss.

4.4 Limitations and Future Research Directions

Several limitations of the current research should be acknowledged:

1. Limited species coverage: While this review covers multiple commercially important species, many cultured species lack detailed longitudinal studies of edible parts development.

2. Environmental interactions: Culture conditions (temperature, water quality, stocking density) interact with duration effects in complex ways not fully addressed in current research.

3. Genetic variation: Heritability estimates indicate significant genetic variation in fillet traits, suggesting potential for selective breeding to optimize the durationedible parts relationship.

Future research should focus on:

Developing predictive models that integrate duration, environmental factors, and genetic potential

Conducting longterm studies across full commercial production cycles

Investigating the molecular mechanisms regulating nutrient partitioning between edible tissues and reproductive structures

Economic analysis of optimal harvest timing considering both yield quantity and quality premiums

5. Conclusion

This study confirms a positive correlation between fish culture duration and the quantity of edible parts, but demonstrates that this relationship is nonlinear, speciesspecific, and subject to multiple interacting factors. Key conclusions include:

1. Fillet weight and body weight show strong positive genetic correlations (>0.98) in Nile tilapia, indicating that prolonged culture within optimal growing periods increases edible yield .

2. Nutritional quality, particularly protein content and EPA+DHA levels, continues to improve with age even after somatic growth slows, as demonstrated in 6yearold Chinese softshelled turtles .

3. Excessive culture duration can reduce edible parts quantity and quality, as shown by declining meat yield and nutritional parameters in crabs held beyond 60 days .

4. The optimal harvest time represents a balance between yield quantity, nutritional quality, and economic efficiency, and varies significantly among species and production systems.

These findings provide scientific justification for moving beyond simple assumptions about "longer is better" in aquaculture, toward evidencebased harvest timing decisions that maximize both production efficiency and product quality.

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Acknowledgments

The author thanks the researchers whose published work formed the foundation of this review, particularly the teams at Southwest University, the Federal University DutsinMa, and the Jiangsu Academy of Agricultural Sciences for their valuable contributions to understanding the relationship between aquatic animal age and edible quality.

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Submitted: Fer 2026